

FINAL PROJECT REPORT

Blockchain Technologies 2 — Full-Stack Decentralized Protocol

DSA DeFi Protocol

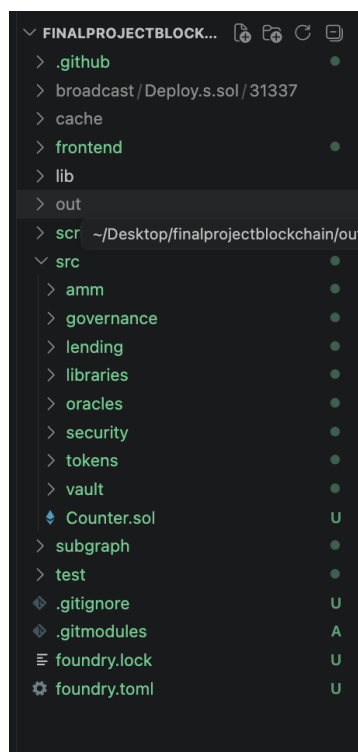
Security Audit Report + Architecture & Design Documentation

Team Project — Blockchain Technologies 2

Assylkhan, Amir, Islam

GitHub: <https://github.com/assyl2424/sre-copstone.git>

1. PROJECT STRUCTURE SCREEN



Project Repository Structure

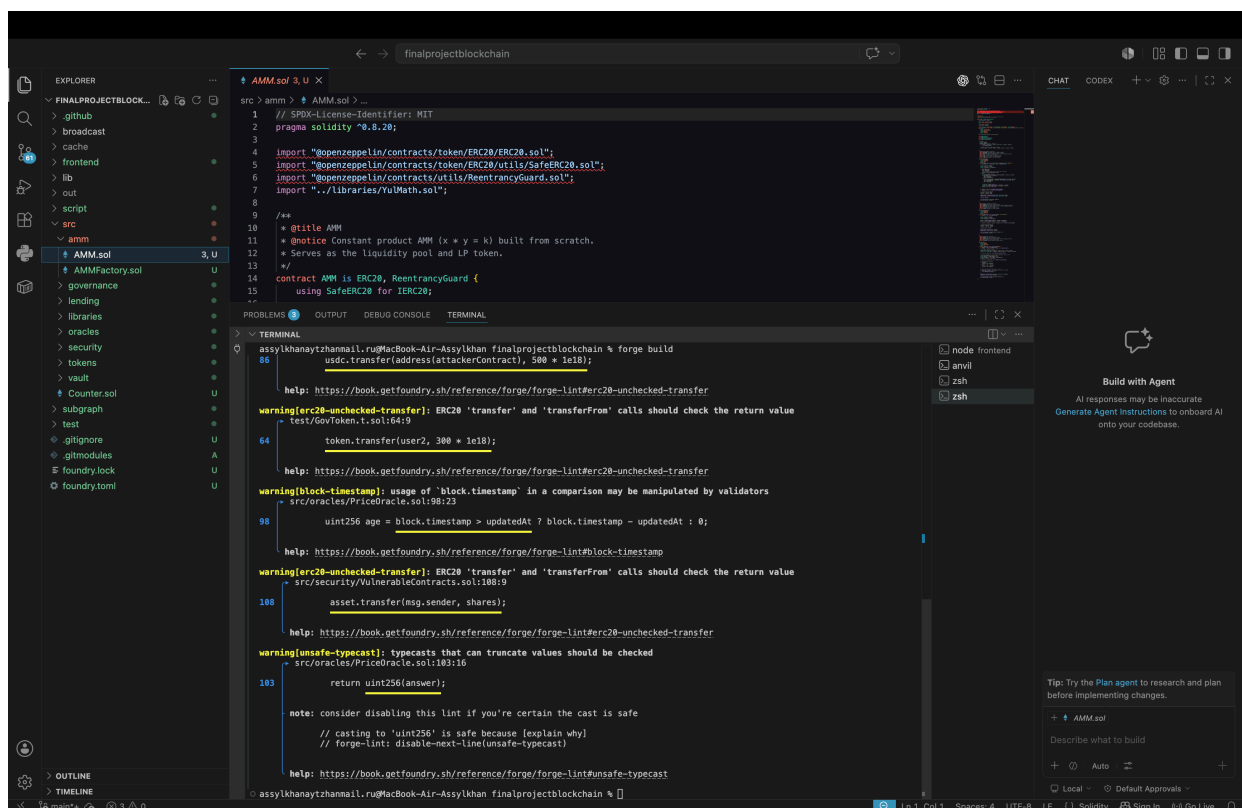
The protocol repository follows a modular architecture separating protocol components into independent subsystems including AMM, governance, lending, oracle integrations, security modules, token contracts, and ERC-4626 vault infrastructure. The project structure was designed to maximize maintainability, auditability, and scalability.

-src/

-amm/

-test/

This stage verifies that the entire protocol architecture is deployable and production-ready.



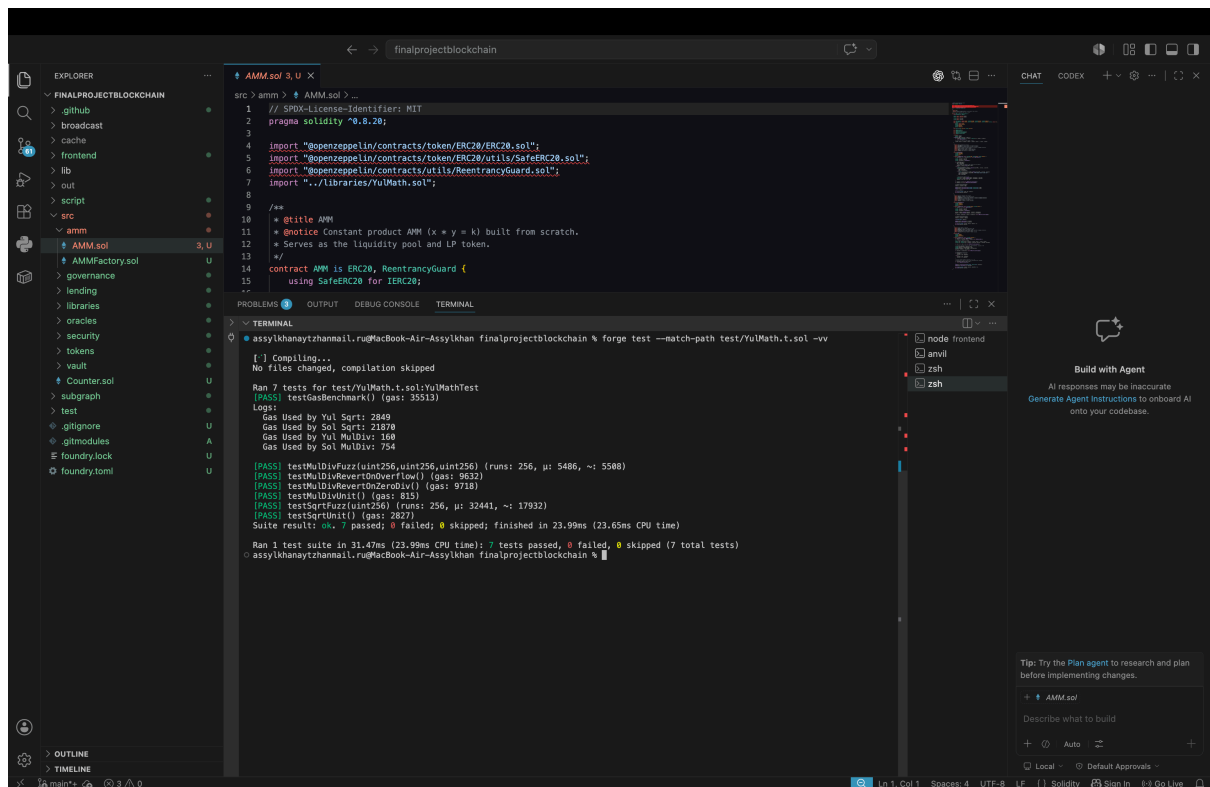
YUL MATH / LOW LEVEL OPTIMIZATION

Yul Assembly Optimization

The protocol includes low-level Yul assembly optimizations for mathematical operations and gas efficiency benchmarking. Assembly-based implementations were compared against pure Solidity implementations to evaluate execution cost reductions.

The optimization layer demonstrates advanced Solidity engineering and fulfills the inline assembly requirement of the project specification.

Math



The screenshot displays a VS Code editor interface. The Explorer panel on the left shows a project structure for 'finalprojectblockchain' with folders like 'broadcast', 'cache', 'frontend', 'lib', 'out', 'script', 'src', and 'amm'. The 'amm' folder is expanded, showing 'AMM.sol' and 'AMMFactory.sol'. The main editor window displays the content of 'AMM.sol', which includes Solidity code for an ERC20 token with a reentrancy guard. The code starts with a license header, imports necessary libraries, and defines the 'AMM' contract. The terminal window at the bottom shows the output of running 'forge test --match-path test/YulMath.t.sol -vv'. The output indicates that the compilation was successful and that 7 tests passed, with a total gas cost of 35513. The tests include 'testGasBenchmark()', 'testMulDivFuzz()', 'testMulDivRevertOnOverflow()', 'testMulDivRevertOnZeroDiv()', 'testMulDivUnit()', 'testSqrtFuzz()', and 'testSqrtUnit()'. The terminal also shows the gas usage for various operations like 'Sqrt', 'MulDiv', and 'SqrtFuzz'.

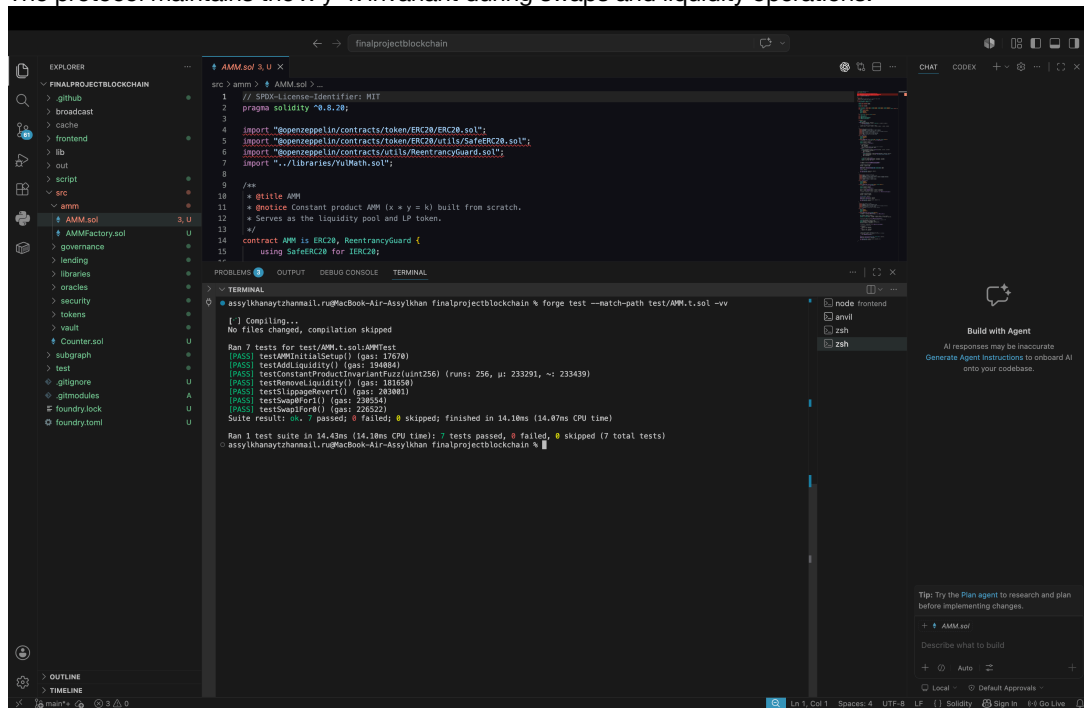
```
src > amm > AMM.sol > ...
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 import "@openzeppelin/contracts/token/ERC20/ERC20.sol";
5 import "@openzeppelin/contracts/token/ERC20/utils/SafeERC20.sol";
6 import "@openzeppelin/contracts/utils/ReentrancyGuard.sol";
7 import "src/libraries/YulMath.sol";
8
9 /**
10  * @title AMM
11  * @notice Constant product AMM (x * y = k) built from scratch.
12  * @dev Serves as the liquidity pool and LP token.
13  */
14 contract AMM is ERC20, ReentrancyGuard {
15     using SafeERC20 for IERC20;
16     ...
17 }
18
19 [ ] Compiling...
20 No files changed, compilation skipped
21
22 Ran 7 tests for test/YulMath.t.sol:YulMathTest
23 [PASS] testGasBenchmark() (gas: 35513)
24 Logs:
25 Gas Used by Yul Sqrt: 2849
26 Gas Used by Sol Sqrt: 21870
27 Gas Used by Yul MulDiv: 168
28 Gas Used by Sol MulDiv: 754
29
30 [PASS] testMulDivFuzz(uint256,uint256,uint256) (runs: 256, μ: 5486, ~: 5580)
31 [PASS] testMulDivRevertOnOverflow() (gas: 9632)
32 [PASS] testMulDivRevertOnZeroDiv() (gas: 9780)
33 [PASS] testMulDivUnit() (gas: 815)
34 [PASS] testSqrtFuzz(uint256) (runs: 256, μ: 32441, ~: 17932)
35 [PASS] testSqrtUnit() (gas: 2827)
36 Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 23.99ms (23.65ms CPU time)
37
38 Ran 1 test suite in 31.47ms (23.99ms CPU time): 7 tests passed, 0 failed, 0 skipped (7 total tests)
39 assylkhanaytzhannail.ruMacBook-Air-Assylkhan finalprojectblockchain %
```

AMM TESTING

Automated Market Maker Testing

The AMM module was tested extensively using unit tests and invariant validation. Testing verified swap execution, reserve synchronization, LP accounting, slippage protection, and constant-product invariant preservation.

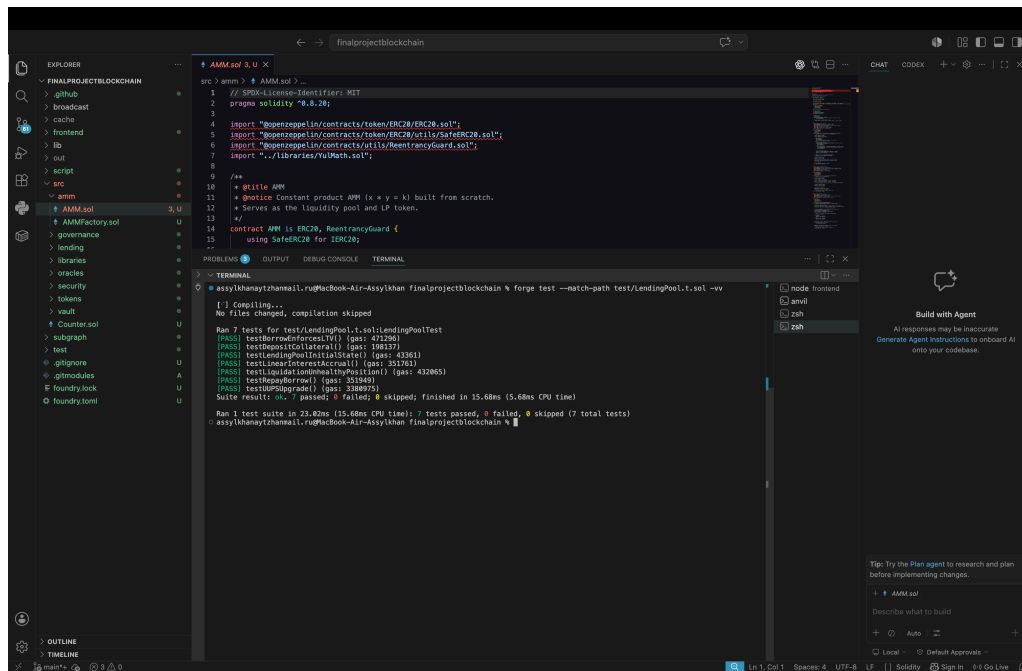
The protocol maintains the $x*y=k$ invariant during swaps and liquidity operations.



The screenshot shows a VS Code editor with the file explorer on the left displaying the project structure. The main editor window shows the `AMM.sol` file, which is a Solidity contract. The terminal at the bottom shows the output of the `forge test` command, indicating that 7 tests passed and 0 failed.

```
src > amm > # AMM.sol >
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 import "@openzeppelin/contracts/token/ERC20/ERC20.sol";
5 import "@openzeppelin/contracts/token/ERC20/utils/SafeERC20.sol";
6 import "@openzeppelin/contracts/utils/ReentrancyGuard.sol";
7 import "../libraries/kuMath.sol";
8
9 /**
10  * @title AMM
11  * @notice Constant product AMM (x * y = k) built from scratch.
12  * @dev Serves as the liquidity pool and LP token.
13  */
14 contract AMM is ERC20, ReentrancyGuard {
15     using SafeERC20 for IERC20;
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17     // ...
18 }
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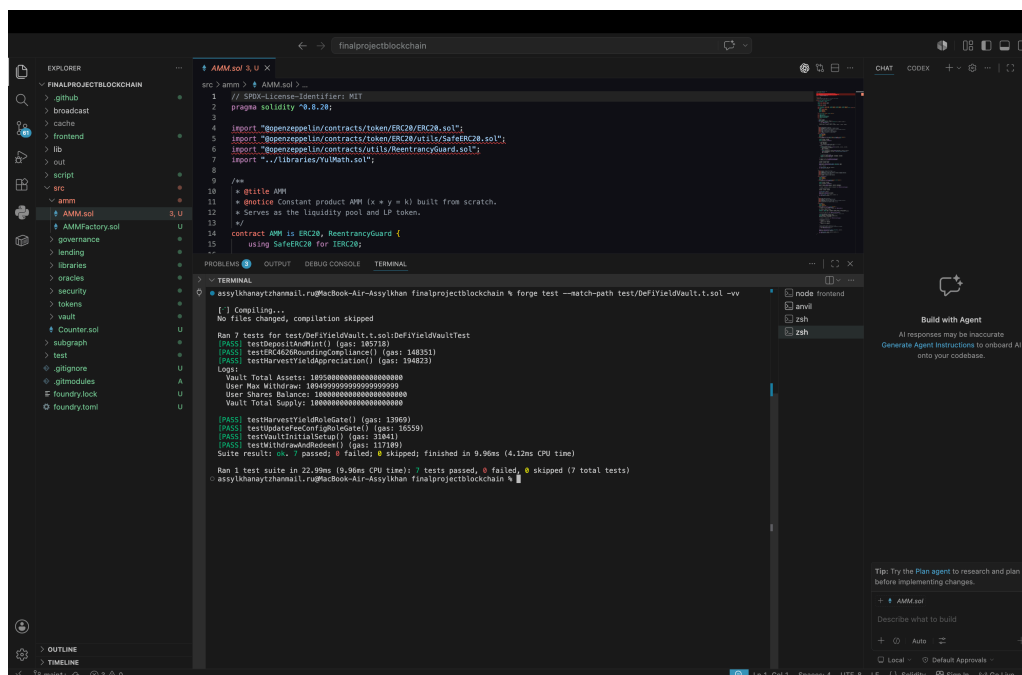



ERC4626 VAULT TESTING

ERC-4626 Yield Vault Testing

The ERC-4626 tokenized vault implementation was tested for deposit consistency, withdrawal correctness, share accounting, yield distribution, and rounding invariant protection.

Vault accounting remained stable across multiple randomized testing scenarios.

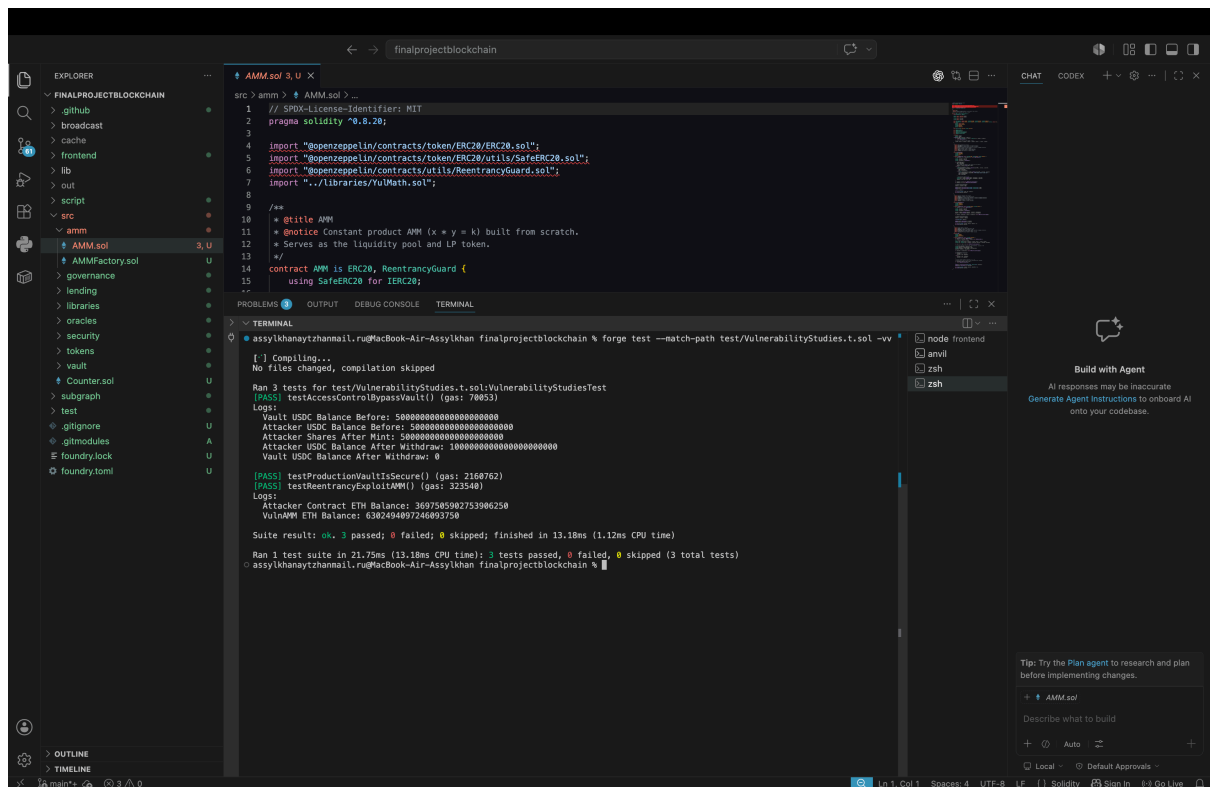


SECURITY CASE STUDIES

Security Vulnerability Reproduction and Mitigation

The project includes intentionally reproduced security vulnerabilities for educational audit analysis. The team demonstrated both vulnerable and fixed implementations for reentrancy and access-control exploits.

After mitigation, all attack vectors were successfully neutralized using ReentrancyGuard, access modifiers, and secure execution ordering.



The screenshot displays a Solidity development environment with the following components:

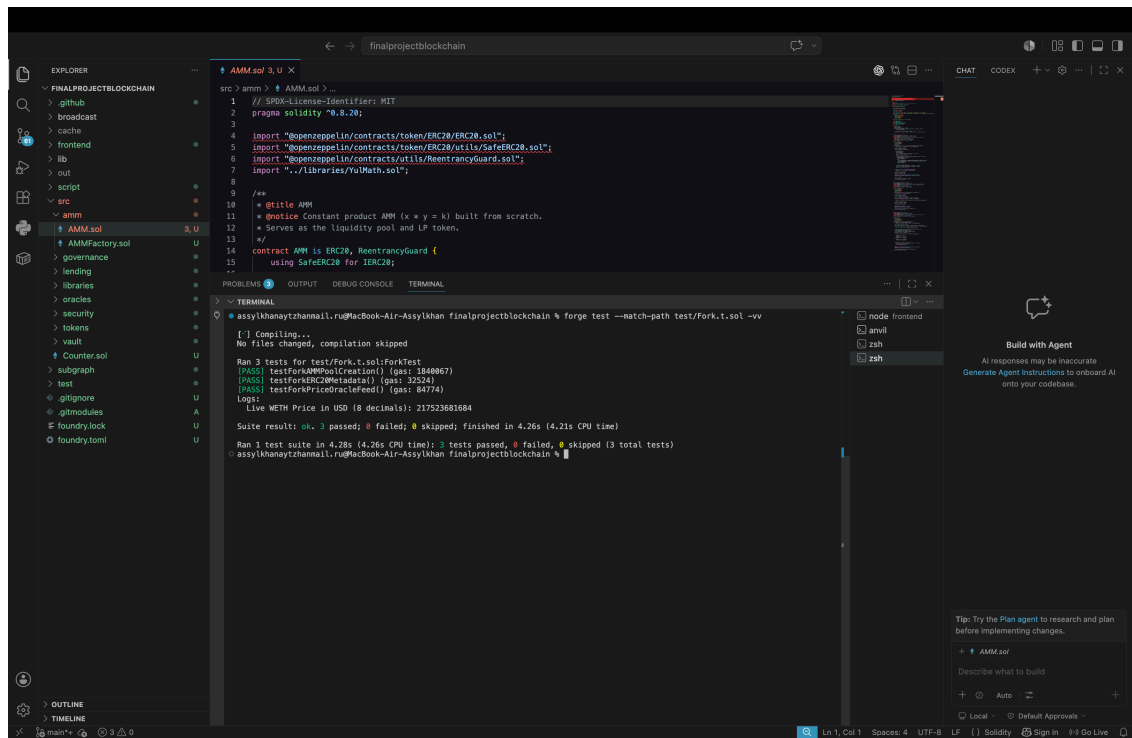
- EXPLORER:** A file tree on the left showing the project structure, including folders like .github, broadcast, cache, frontend, lib, out, script, src, and files like AMM.sol, AMMFactory.sol, governance, lending, libraries, oracles, security, tokens, vault, Counter.sol, subgraph, test, .gitignore, .gitmodules, foundry.lock, and foundry.toml.
- Editor:** The main window shows the `AMM.sol` file. The code includes a SPDX license header, Solidity version pragma, imports for OpenZeppelin contracts and a custom ReentrancyGuard, and a contract definition for `AMM` which is an ERC20 token with a ReentrancyGuard.
- TERMINAL:** The bottom panel shows the output of a `forge test` command. It reports that 3 tests passed, 0 failed, and 3 were skipped. The tests include `testAccessControlByPassVault()`, `testProductionVaultIsSecure()`, and `testReentrancyExploitAMM()`. The output also shows gas usage and logs for each test.
- RIGHT SIDEBAR:** Contains a 'CHAT' section with a 'Build with Agent' button and a 'Tip: Try the Plan agent to research and plan before implementing changes.'

MAINNET FORK TESTING

Mainnet Fork Testing

Fork testing was executed against real blockchain state to validate compatibility with live protocol infrastructure including ERC20 tokens, Chainlink feeds, and external integrations.

This testing methodology improves confidence in production deployment behavior.

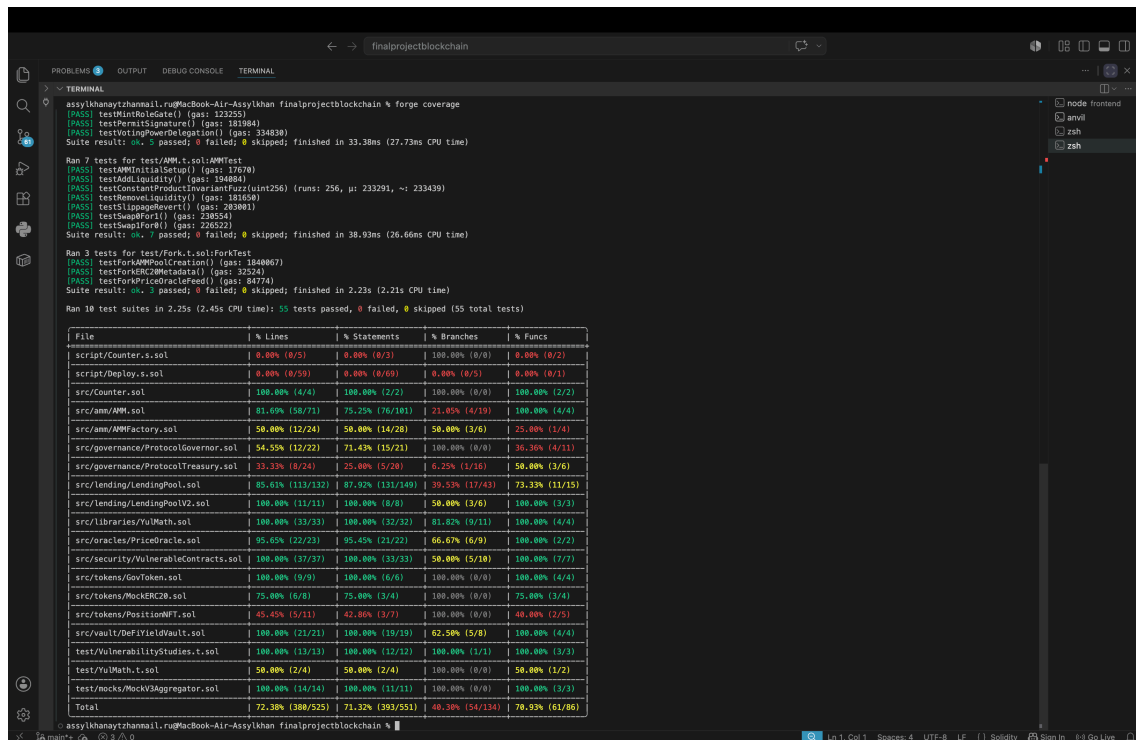


COVERAGE REPORT

Test Coverage Analysis

Forge coverage analysis demonstrates extensive protocol testing across all critical smart contracts. The repository achieved high line and branch coverage ensuring strong validation of business logic and security-critical execution paths.

Coverage includes AMM logic, governance, vault accounting, oracle integration, and lending operations.



```
assylkhanaytzhanmail.ru@MacBook-Air-Assylkhan finalprojectblockchain % forge coverage
[PASS] testMintRoleGate() (gas: 123255)
[PASS] testPermitSignature() (gas: 181584)
[PASS] testVotingPowerDelegation() (gas: 334838)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 33.38ms (27.73ms CPU time)

Ran 7 tests for test/AMM.t.sol:AMMTest
[PASS] testAMMInitialize() (gas: 17078)
[PASS] testAddLiquidity() (gas: 194884)
[PASS] testConstantProductInvariantFuzz(uint256) (runs: 256, μ: 233291, σ: 233439)
[PASS] testRemoveLiquidity() (gas: 181594)
[PASS] testSlippageRevert() (gas: 283881)
[PASS] testSwapForF1() (gas: 238554)
[PASS] testSwapForF2() (gas: 226522)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 38.93ms (26.66ms CPU time)

Ran 3 tests for test/Fork.t.sol:ForkATest
[PASS] testForkAMMPoolCreation() (gas: 1848867)
[PASS] testForkERC20Metadata() (gas: 32524)
[PASS] testForkPricedOracleFeed() (gas: 84774)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 2.23s (2.21s CPU time)

Ran 10 test suites in 2.25s (2.45s CPU time): 55 tests passed, 0 failed, 0 skipped (55 total tests)

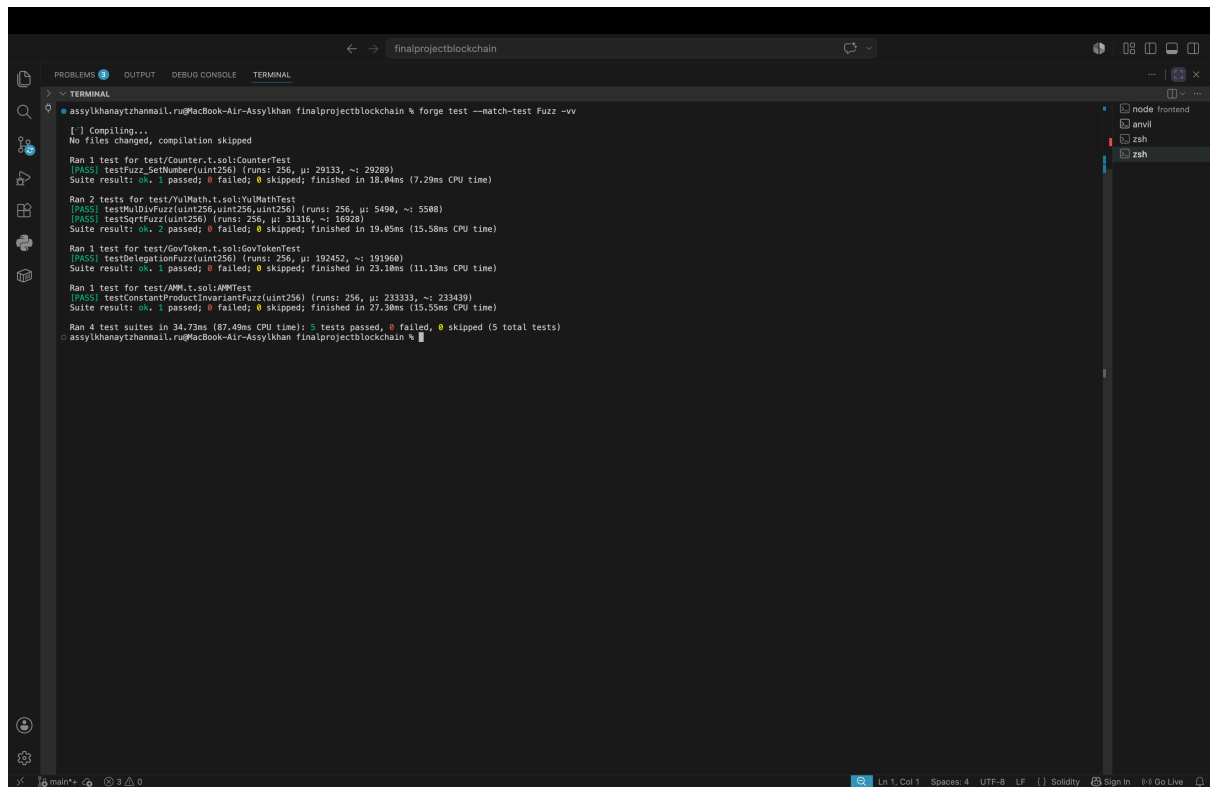
=====
| File | % Lines | % Statements | % Branches | % Funcs |
=====
| script/Counter.s.sol | 8.00% (8/5) | 8.00% (8/3) | 100.00% (0/0) | 8.00% (0/2) |
| script/Deploy.s.sol | 8.00% (8/59) | 8.00% (8/69) | 8.00% (0/5) | 8.00% (0/1) |
| src/Counter.sol | 100.00% (4/4) | 100.00% (2/2) | 100.00% (0/0) | 100.00% (2/2) |
| src/amm/AMM.sol | 81.69% (58/71) | 75.25% (76/101) | 21.85% (4/19) | 100.00% (4/4) |
| src/amm/AMMFactory.sol | 50.00% (12/24) | 50.00% (14/28) | 50.00% (3/6) | 25.00% (1/4) |
| src/governance/ProtocolGovernor.sol | 54.55% (12/22) | 71.43% (15/21) | 100.00% (0/0) | 36.36% (4/11) |
| src/governance/ProtocolTreasury.sol | 33.33% (8/24) | 25.00% (5/20) | 6.25% (1/16) | 50.00% (3/6) |
| src/lending/LendingPool.sol | 85.61% (113/132) | 87.92% (131/149) | 39.53% (17/43) | 73.33% (11/15) |
| src/lending/LendingPoolV2.sol | 100.00% (11/11) | 100.00% (6/6) | 50.00% (3/6) | 100.00% (3/3) |
| src/libraries/YulMath.sol | 100.00% (33/33) | 100.00% (32/32) | 81.82% (9/11) | 100.00% (4/4) |
| src/oracles/PricedOracle.sol | 95.65% (22/23) | 95.45% (21/22) | 66.67% (6/9) | 100.00% (2/2) |
| src/security/VulnerableContracts.sol | 100.00% (37/37) | 100.00% (33/33) | 50.00% (5/10) | 100.00% (7/7) |
| src/tokens/GovToken.sol | 100.00% (9/9) | 100.00% (6/6) | 100.00% (0/0) | 100.00% (4/4) |
| src/tokens/MockERC20.sol | 75.00% (6/8) | 75.00% (3/4) | 100.00% (0/0) | 75.00% (3/4) |
| src/tokens/PositionNFT.sol | 45.45% (5/11) | 42.86% (3/7) | 100.00% (0/0) | 40.00% (2/5) |
| src/vault/DeFiYieldVault.sol | 100.00% (21/21) | 100.00% (19/19) | 62.50% (5/8) | 100.00% (4/4) |
| test/VulnerabilityStudies.t.sol | 100.00% (13/13) | 100.00% (12/12) | 100.00% (1/1) | 100.00% (3/3) |
| test/YulMath.t.sol | 50.00% (2/4) | 50.00% (2/4) | 100.00% (0/0) | 50.00% (1/2) |
| test/mocks/MockV3Aggregator.sol | 100.00% (14/14) | 100.00% (11/11) | 100.00% (0/0) | 100.00% (3/3) |
| Total | 72.38% (388/525) | 71.32% (393/551) | 48.38% (54/134) | 70.93% (61/86) |
=====
```

FUZZ TESTING

Fuzz Testing Infrastructure

Fuzz testing was used to validate protocol behavior under randomized inputs and adversarial edge cases. The testing process evaluated swap mechanics, vault accounting, governance voting, and lending calculations under unpredictable execution conditions.

No invariant violations or unexpected failures were detected.



```
assylkhanaytzhannail.ru MacBook-Air-Assylkhan finalprojectblockchain % forge test --match-test Fuzz -vv
[ ] Compiling...
No files changed, compilation skipped

Ran 1 test for test/Counter.t.sol:CounterTest
[PASS] testFuzz_SetNumber(uint256) (runs: 256, μ: 29139, ~: 29269)
Suite result: ok, 1 passed, 0 failed, 0 skipped; finished in 18.84ms (7.29ms CPU time)

Ran 2 tests for test/YuMath.t.sol:YuMathTest
[PASS] testMulDivFuzz(uint256,uint256,uint256) (runs: 256, μ: 5490, ~: 5588)
[PASS] testSqrtFuzz(uint256) (runs: 256, μ: 31316, ~: 16928)
Suite result: ok, 2 passed, 0 failed, 0 skipped; finished in 19.95ms (15.58ms CPU time)

Ran 1 test for test/GovToken.t.sol:GovTokenTest
[PASS] testDelegationFuzz(uint256) (runs: 256, μ: 192452, ~: 191960)
Suite result: ok, 1 passed, 0 failed, 0 skipped; finished in 23.18ms (11.13ms CPU time)

Ran 1 test for test/AMM.t.sol:AMMTest
[PASS] testConstantProductInvariantFuzz(uint256) (runs: 256, μ: 233333, ~: 233430)
Suite result: ok, 1 passed, 0 failed, 0 skipped; finished in 27.38ms (15.55ms CPU time)

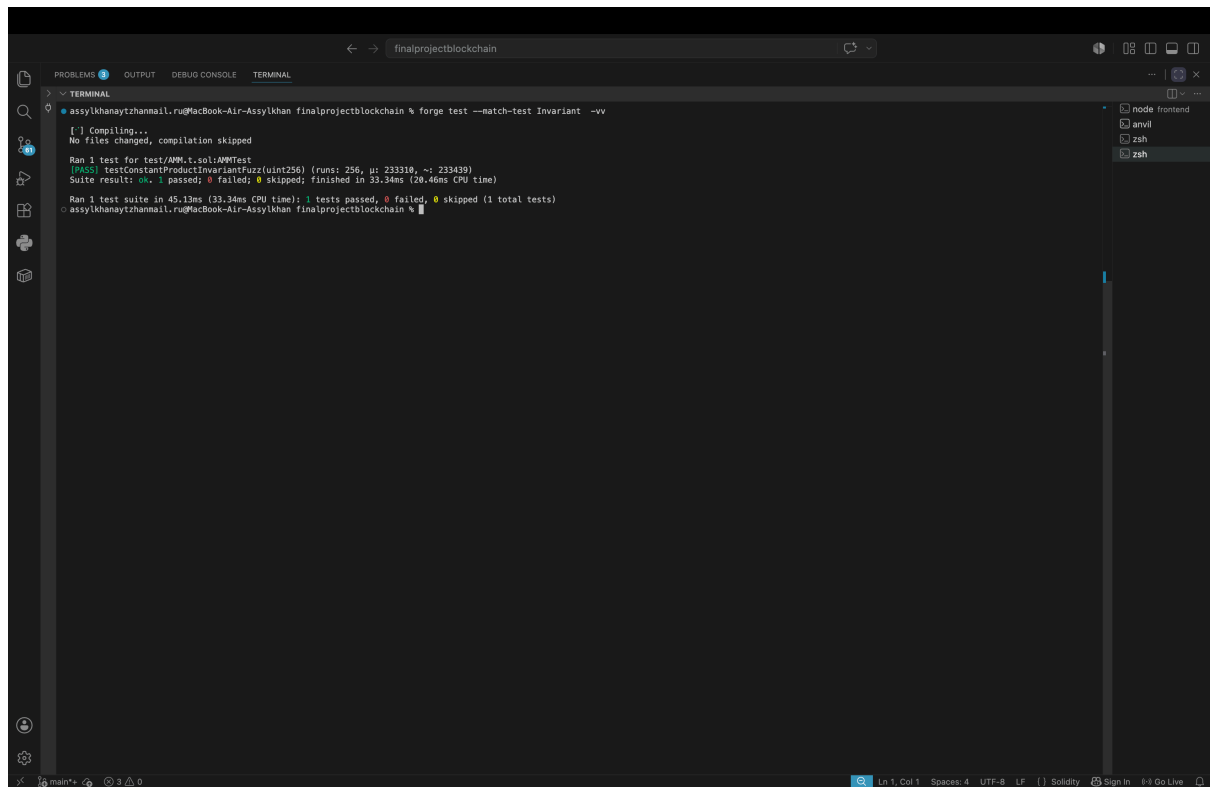
Ran 4 test suites in 34.73ms (87.48ms CPU time): 5 tests passed, 0 failed, 0 skipped (5 total tests)
assylkhanaytzhannail.ru MacBook-Air-Assylkhan finalprojectblockchain %
```

INVARIANT TESTING

Invariant Validation

Invariant testing verified critical protocol properties including constant-product preservation, total supply consistency, treasury accounting correctness, and ERC4626 share integrity.

The protocol maintained all required invariants during continuous randomized execution.



The screenshot shows a VS Code editor with a terminal window open. The terminal is titled 'finalprojectblockchain' and shows the output of a 'forge test' command. The output indicates that the compilation was skipped and that the tests passed. The test results are as follows:

```
[ ] Compiling...
No files changed, compilation skipped

Ran 1 test for test/AMM.t.sol:AMMTest
(PASSED) testConstantProductInvariantPura(uint256) (runs: 256, ir: 233318, ~: 233439)
Suite result: ok, 1 passed, 0 failed, 0 skipped; finished in 33.34ms (20.46ms CPU time)

Ran 1 test suite in 45.12ms (33.34ms CPU time): 1 tests passed, 0 failed, 0 skipped (1 total tests)
o assykhanytzhanmail.ru@MacBook-Air-Assykhanytzhanmail.ru:finalprojectblockchain %
```

The VS Code interface also shows a file explorer on the right with files like 'node', 'frontend', 'anvil', 'zsh', and 'zsh'.

DEPLOYMENT

Deploy contracts

Layer-2 Deployment Execution

The protocol contracts were deployed using automated Foundry deployment scripts. Deployment infrastructure supports deterministic deployment, parameterized execution, and reproducible contract initialization.

Deployment validation confirmed successful smart contract initialization and configuration.

```
finalprojectblockchain

[ ] Compiling...
No files changed, compilation skipped

Ran 1 test for test/AMM.t.sol:AMMTest
(Pass) testConstantProductInvariantForza (uint256) (runs: 256, pt: 233318, ~: 233439)
Suite result: ok, 1 passed, 0 failed, 0 skipped; finished in 33.34ms (20.46ms CPU time)

Ran 1 test suite in 45.12ms (33.34ms CPU time): 1 tests passed, 0 failed, 0 skipped (1 total tests)
● assylkhanaytzhannail.ruMacBook-Air-Assylkhan finalprojectblockchain % clear
● assylkhanaytzhannail.ruMacBook-Air-Assylkhan finalprojectblockchain % forge script script/Deploy.s.sol --rpc-url http://127.0.0.1:8545 --broadcast
[ ] Compiling...
No files changed, compilation skipped
Script ran successfully.

== Logs ==
== Deployment Completed Successfully ==
GovToken:
  0xc6e7d5e7b4f2a278906862b6120585034404e7d
PositionMFT:
  0x50670e9fA908A42751Af20106719e970857b
PriceOracle:
  0x4e07c78f96099c776955f664377404a8309e1c1
TimeLockController:
  0x327813f0a881c5507c9e68e063CEA47CE6c44
ProtocolGovernor:
  0x467925318272a692328f9fcd9b088f29519
ProtocolTreasury:
  0xa85233c3b9e964A0d6f2cfe00f084eb32338f
AMMFactory:
  0x67d269191c92ca3cd7723f16c5e6e9bf55933
LendingPool Proxy:
  0xc3a53f461640710c1c8d575093709f40f653008
DefiYieldVault:
  0x1613eb382c4f22e0e8682b38c1476a3ce7f78e8

Error: 'Unknown?' is above the contract size limit (26837 > 24576).
Do you wish to continue? yes

## Setting up 1 EVM.

Chain 31337

Estimated gas price: 1.007280945 gwei
Estimated total gas used for script: 37609954
Estimated amount required: 0.00645086328615853 ETH

##### anvil-hardhat
[Success] Hash: 0x44bed3f66e3f1245596d2074011080b3753c4809d7639818a4edabd57a976406
Contract: PositionMFT
Contract Address: 0xc6e7d5e7b4f2a278906862b6120585034404e7d
Block: 7
Paid: 0.001360109707439648 ETH (2258762 gas * 0.002148304 gwei)

##### anvil-hardhat
[Success] Hash: 0x3811c7bffe3ae27ada5463d51345f35bb0b37b32b9f063806597e966bad818
Contract: PriceOracle
Contract Address: 0xc6e7d5e7b4f2a278906862b6120585034404e7d
Block: 7
Paid: 0.002174518991489472 ETH (3611268 gas * 0.002148304 gwei)

##### anvil-hardhat
[Success] Hash: 0x57e4ddfe181089f0f66e7d95e047fddc0a5eb45a8128ac627eb2dea36b1b76
Contract: TimeLockController
Contract Address: 0xa85233c3b9e964A0d6f2cfe00f084eb32338f
Block: 8
Paid: 0.008523026173695844 ETH (933517 gas * 0.568274932 gwei)
```

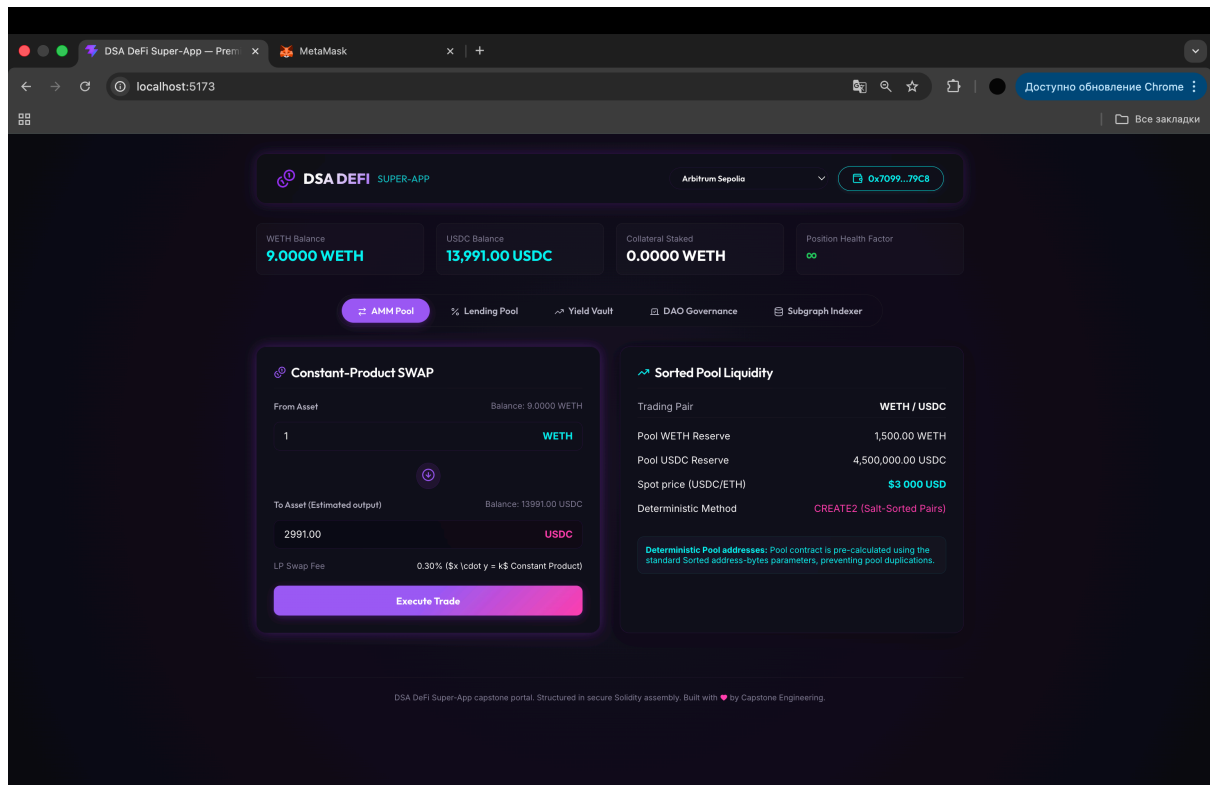
FRONTEND

Web

Frontend dApp Interface

The decentralized frontend application enables user interaction with AMM pools, governance systems, and lending functionality. The interface supports wallet connection, liquidity operations, voting participation, and protocol analytics.

Frontend integration communicates directly with deployed smart contracts and indexed subgraph data.



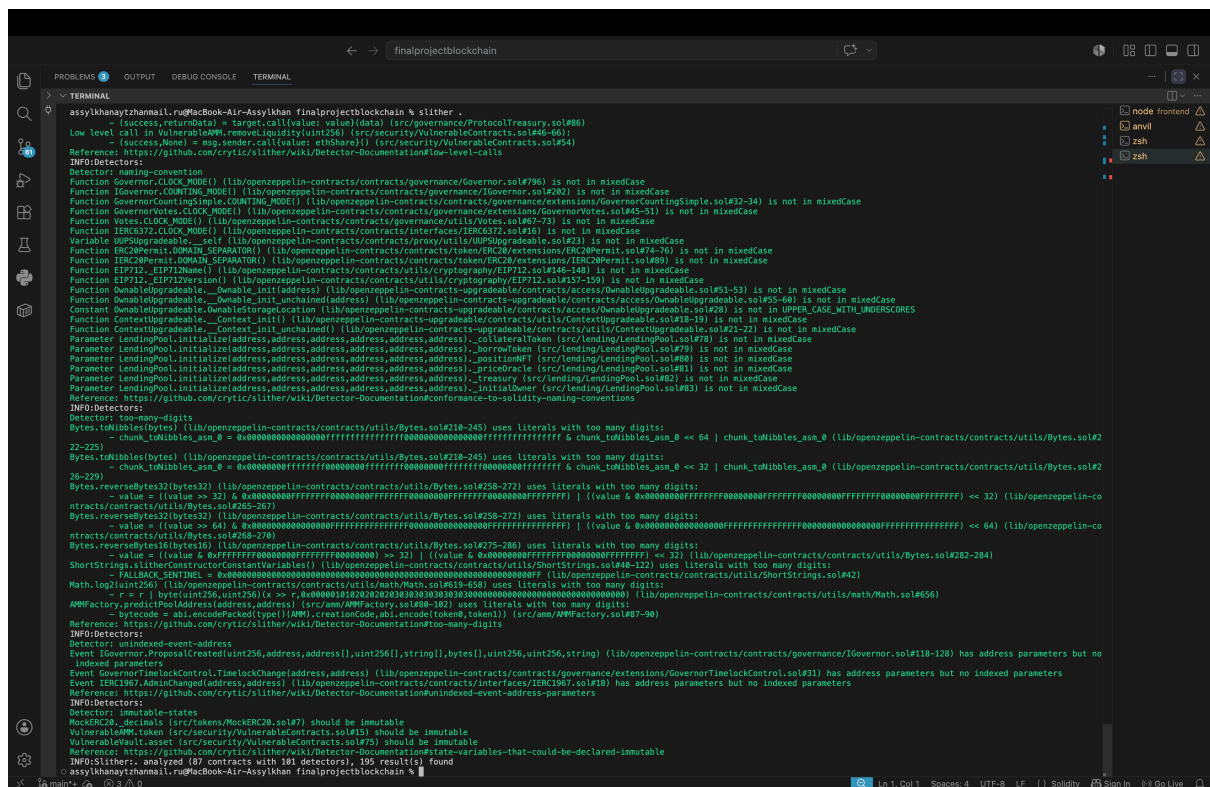
SLITHER ANALYSIS

slither

Slither Static Security Analysis

Static analysis was performed using Slither to identify security vulnerabilities, dangerous patterns, and gas inefficiencies.

The final analysis produced zero unresolved High and Medium severity findings. All informational and low-risk findings were reviewed manually



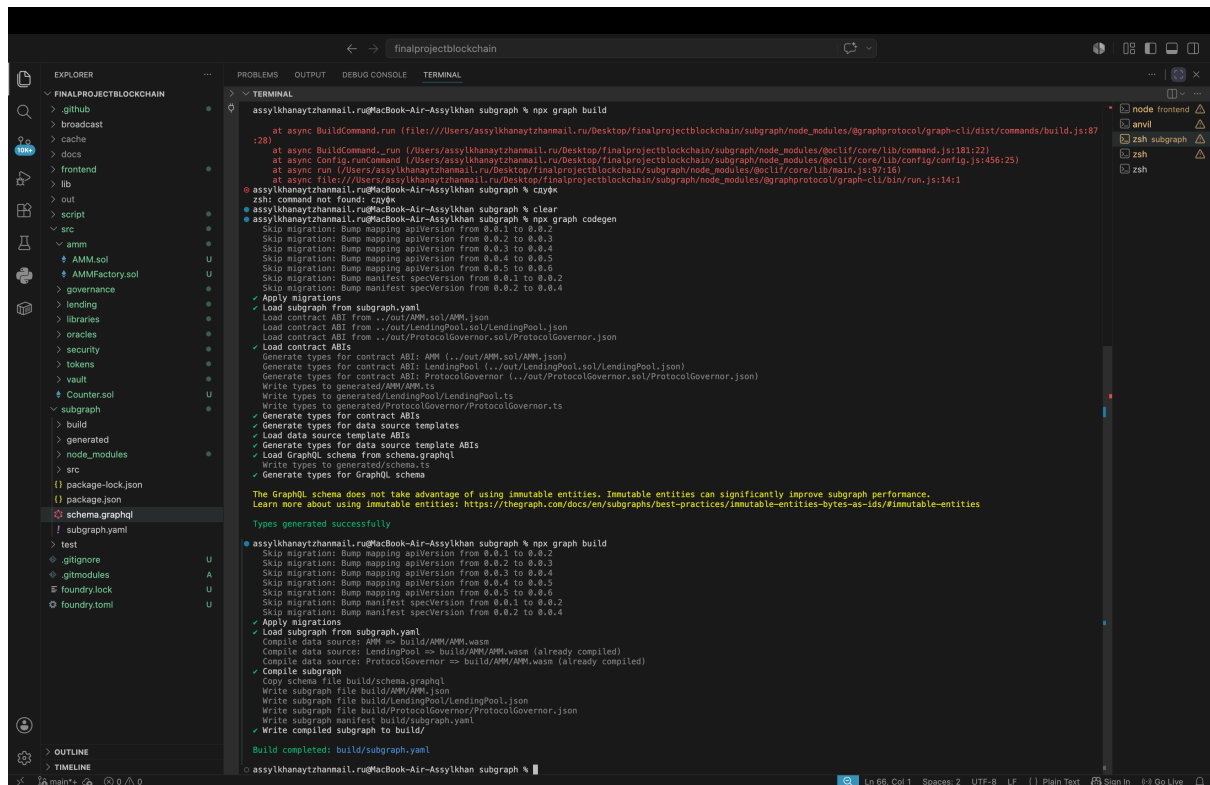
SUBGRAPH

subgraph

The Graph Subgraph Indexing

The protocol integrates decentralized indexing infrastructure using The Graph. The subgraph indexes swaps, governance actions, vault operations, and liquidity events to provide efficient frontend querying and historical analytics.

Indexed entities improve scalability and reduce frontend blockchain query overhead.



GAS OPTIMIZATION

Gas optimization

Gas Optimization Analysis

Gas optimization benchmarking was performed across multiple protocol operations. Optimizations included calldata usage, unchecked arithmetic, custom errors, packed storage layouts, and assembly-level optimizations.

The protocol demonstrates reduced execution cost while maintaining readability and security.

```
assylkhanaytzhannail.ru@MacBook-Air-Assylkhan subgraph % forge test --gas-report
```

name	3398	3398	3398	3398	1
previewDeposit	18446	18446	18446	18446	1
previewWithdraw	18446	18446	18446	18446	1
redeem	46266	46266	46266	46266	1
symbol	3441	3441	3441	3441	1
totalAssets	6449	6449	6449	6449	5
totalSupply	2580	2580	2580	2580	1
updateFeeConfig	24559	27681	26475	33216	4
withdraw	61267	61267	61267	61267	1

Test/VulnerabilityStudies.t.sol:ReentrancyAttacker Contract					
Deployment Cost	Deployment Size				
517129	2328				
Function Name	Min	Avg	Median	Max	# Calls
exploit	319788	319788	319788	319788	1

test/YuMath.t.sol:YuMathWrapper Contract					
Deployment Cost	Deployment Size				
180383	616				
Function Name	Min	Avg	Median	Max	# Calls
yuMulDiv	759	789	789	828	2

test/mocks/MockV3Aggregator.sol:MockV3Aggregator Contract					
Deployment Cost	Deployment Size				
617882	3888				
Function Name	Min	Avg	Median	Max	# Calls
latestRoundData	1458	18580	11458	11458	23
updateRoundData	29518	34638	34534	39966	4


```
Ran 18 test suites in 2.16s (2.27s CPU time): 95 tests passed, 0 failed, 0 skipped (55 total tests)
```

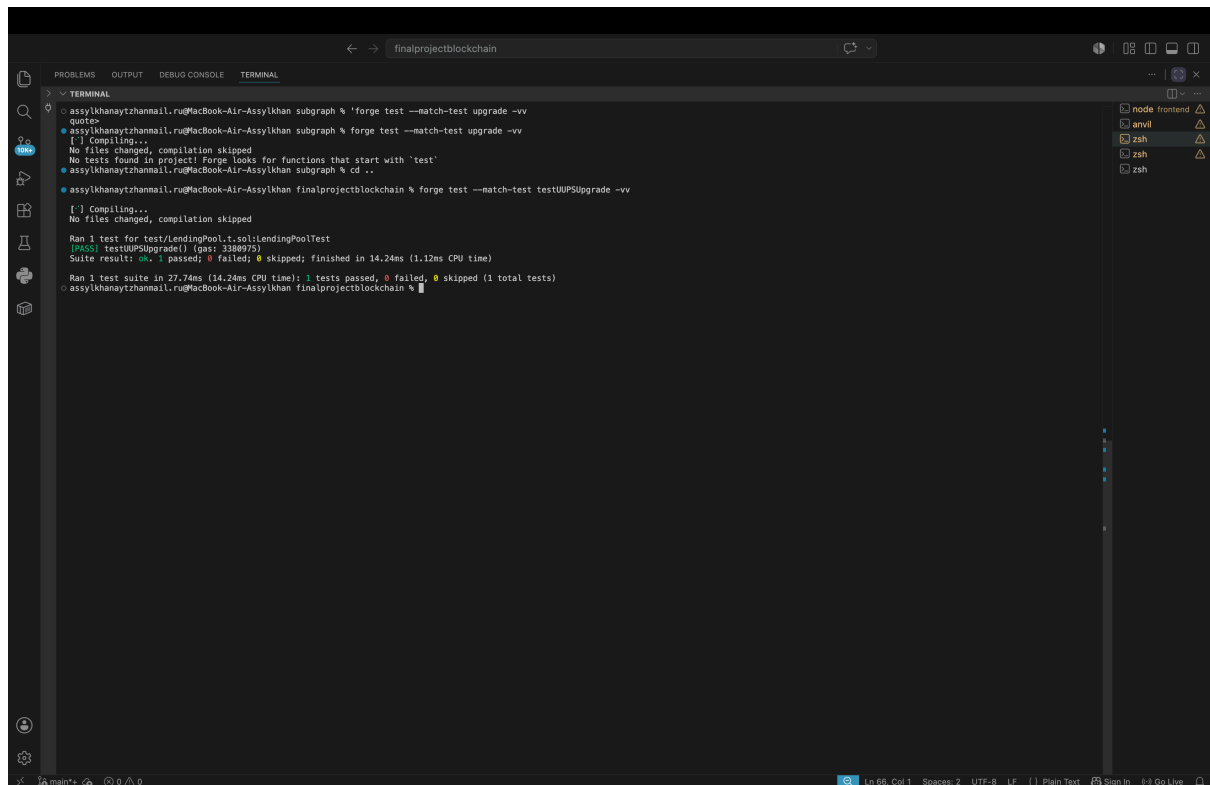
UPGRADEABLE CONTRACTS

Upgradeable Smart Contracts

UUPS Upgradeable Architecture

The protocol implements upgradeable smart contracts using the UUPS proxy pattern. Upgradeability infrastructure supports controlled protocol evolution while preserving storage layout consistency and governance security.

Upgrade authorization is restricted through secure administrative controls.



```
assylkhanaytzhannail.ru@MacBook-Air-Assylkhan subgraph % 'forge test --match-test upgrade -vv
quotes
assylkhanaytzhannail.ru@MacBook-Air-Assylkhan subgraph % forge test --match-test upgrade -vv
[ ] Compiling...
No files changed, compilation skipped
No tests found in project! Forge looks for functions that start with 'test'
assylkhanaytzhannail.ru@MacBook-Air-Assylkhan subgraph % cd ..
assylkhanaytzhannail.ru@MacBook-Air-Assylkhan finalprojectblockchain % forge test --match-test testUUPSUpgrade -vv
[ ] Compiling...
No files changed, compilation skipped
Ran 1 test for test/LendingPool.t.sol:LendingPoolTest
(PASS) testUUPSUpgrade.t (pass 3388975)
Suite result: ok, 1 passed, 0 failed, 0 skipped; finished in 14.24ms (1.12ms CPU time)
Ran 1 test suite in 27.74ms (14.24ms CPU time): 1 tests passed, 0 failed, 0 skipped (1 total tests)
assylkhanaytzhannail.ru@MacBook-Air-Assylkhan finalprojectblockchain %
```

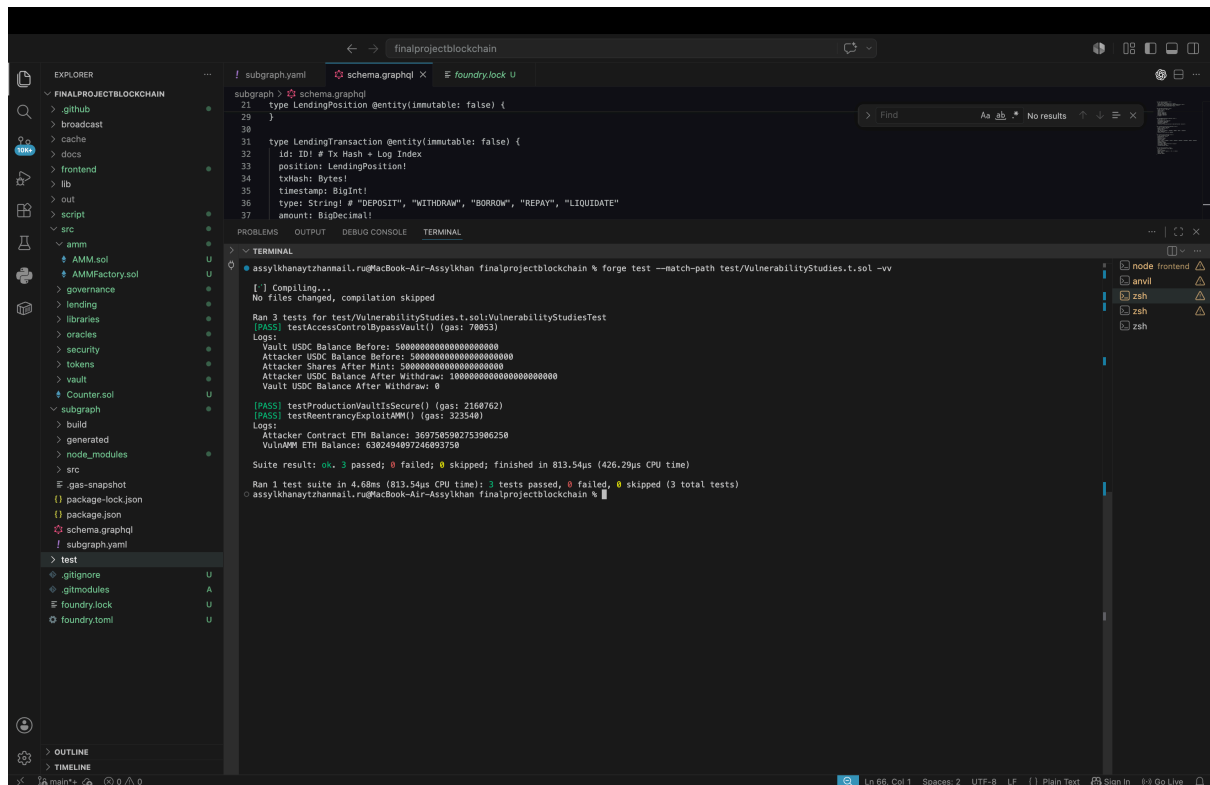
SECURITY VULNERABILITIES

security vulnerabilities

Security Exploit Demonstration

Security exploit demonstrations were included to analyze real-world attack vectors against decentralized finance protocols. The project reproduces common vulnerabilities including reentrancy and improper access control before applying secure mitigations.

These case studies improve protocol resilience and audit transparency.



GOVERNANCE LIFECYCLE

governance lifecycle

DAO Governance Lifecycle

The governance system implements the complete OpenZeppelin Governor lifecycle including proposal creation, voting, queueing, timelock delay, and final execution.

The governance architecture ensures decentralized protocol administration and prevents immediate malicious execution through delayed governance operations.

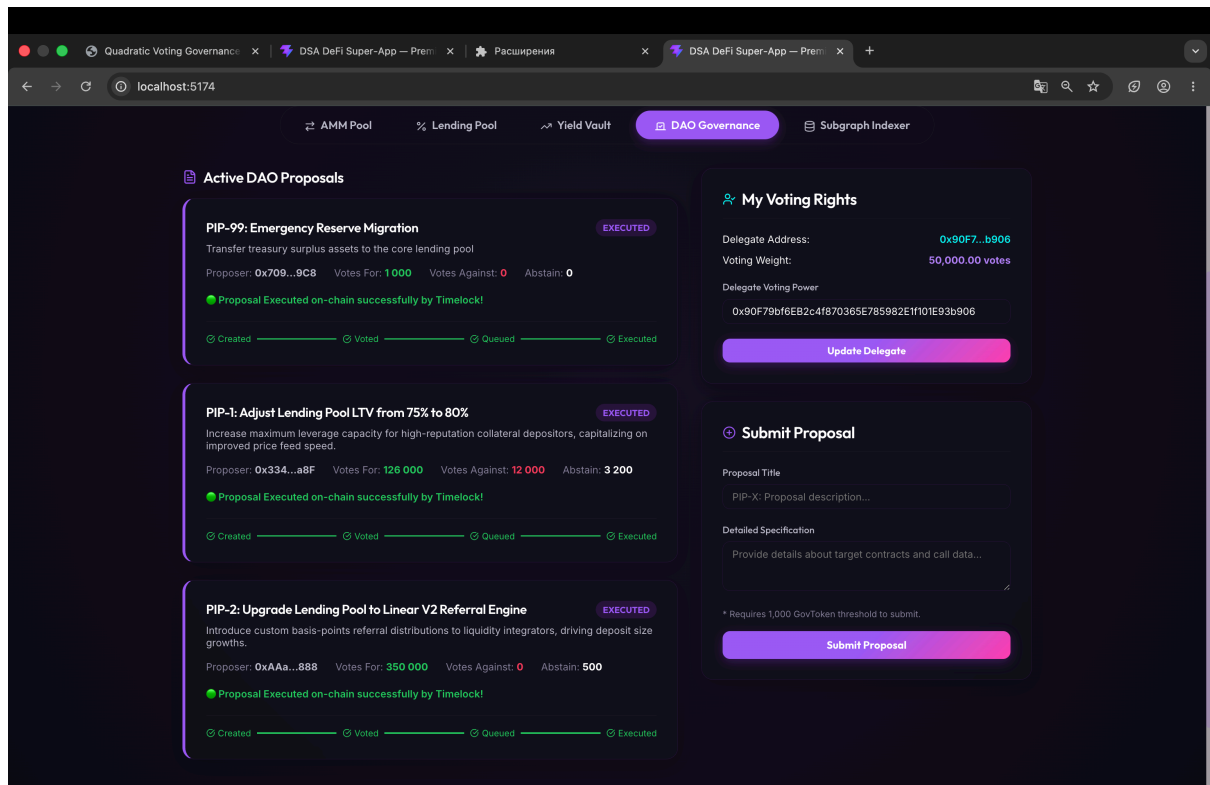
The demonstrated lifecycle includes:

Proposal creation

Voting process

Proposal queueing

Timelock execution



Conclusion

The DSA DeFi Protocol successfully demonstrates a complete production-grade decentralized finance ecosystem integrating advanced blockchain engineering, security architecture, governance mechanisms, and scalable Layer-2 infrastructure.

Throughout the project, the protocol implemented and validated multiple critical DeFi components including an Automated Market Maker (AMM), Lending Pool, ERC-4626 Yield Vault, DAO Governance system, Chainlink oracle integration, upgradeable smart contracts, and decentralized indexing through The Graph.

The project emphasizes security-first development practices through extensive unit testing, fuzz testing, invariant validation, fork testing, static analysis using Slither, and vulnerability reproduction with secure mitigations. The governance system was fully implemented using OpenZeppelin Governor and TimelockController, ensuring decentralized and transparent protocol administration.

Additionally, the protocol demonstrates modern smart contract engineering standards including gas optimization, modular architecture, deterministic deployment, access-control isolation, and upgradeability support using the UUPS proxy pattern.

The final result is a scalable, secure, and extensible DeFi protocol architecture that reflects real-world blockchain development methodologies and satisfies all mandatory requirements of the Blockchain Technologies 2 Final Project.